

# 浙江大学



## Lab Practice Report for Undergraduate

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College: ZheJiang University

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Department: Science

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Major: Marine Science

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Student NO.:

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Instructor: Yanan Di

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Date: 2024.10.10

# Lab Practice Report of Zhejiang University

(此页可在 <http://bksy.zju.edu.cn/office/> 下载)

Course Name: Marine biology Type of Practice: dissection

Name of Practice: Quantitative confirmation of microalgae

Student Name: \_\_\_\_\_ Major: Marine biology

Student Number: \_\_\_\_\_

Team Member: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.10.10

## 1. Aims and Requirement of the Practice (Compulsory)

- (1) Learn about crustacean biology and complete a live dissection of portunidae
- (2) Identify the external anatomy of portunidae; describe the function of important external features.
- (3) Identify the major internal organs of these species and their functions related to swimming, digestion, and respiration.
- (4) Tissue homogenization and preservation for further examination
- (5) Demonstrate dissection skills

## 2. Background information of the Practice

Crustaceans is a Phylum Arthropoda (having exoskeleton and segmented appendages). And penaeidae is a species of Crustaceans.

Class Malacostraca and order Decapoda are two important members of crustaceans. The former is of great importance for aquaculture, with at least 19 segments and order Decapoda having five pairs of pereopods and five pairs of pleopods.

Most decapod crustaceans have a fan-shaped tail called telson, comprised of several uropods.

We can identify sex by looking at the abdominal morphology. The narrower ones are males and the wider ones are females

### 3. Equipment and materials

- (1) .portunidae
- (2) Dissection set
- (3) Ice box and debris ice
- (4) 1.5ml centrifuge tubes and racks
- (5) PBS ,homogenizer and tips,sample preservation box
- (6) Marker pens,magnifying glass
- (7) Gloves etc.

### 4. Procedures

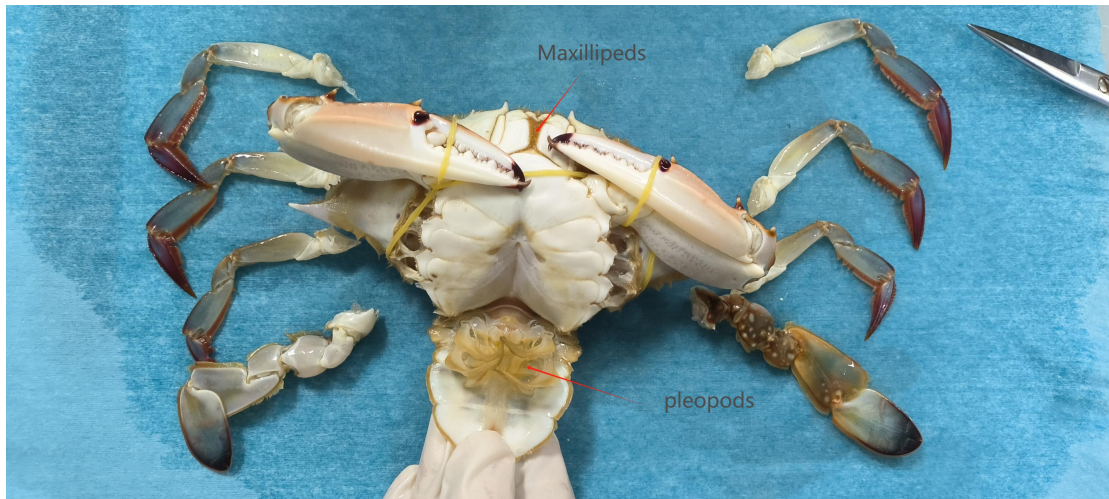
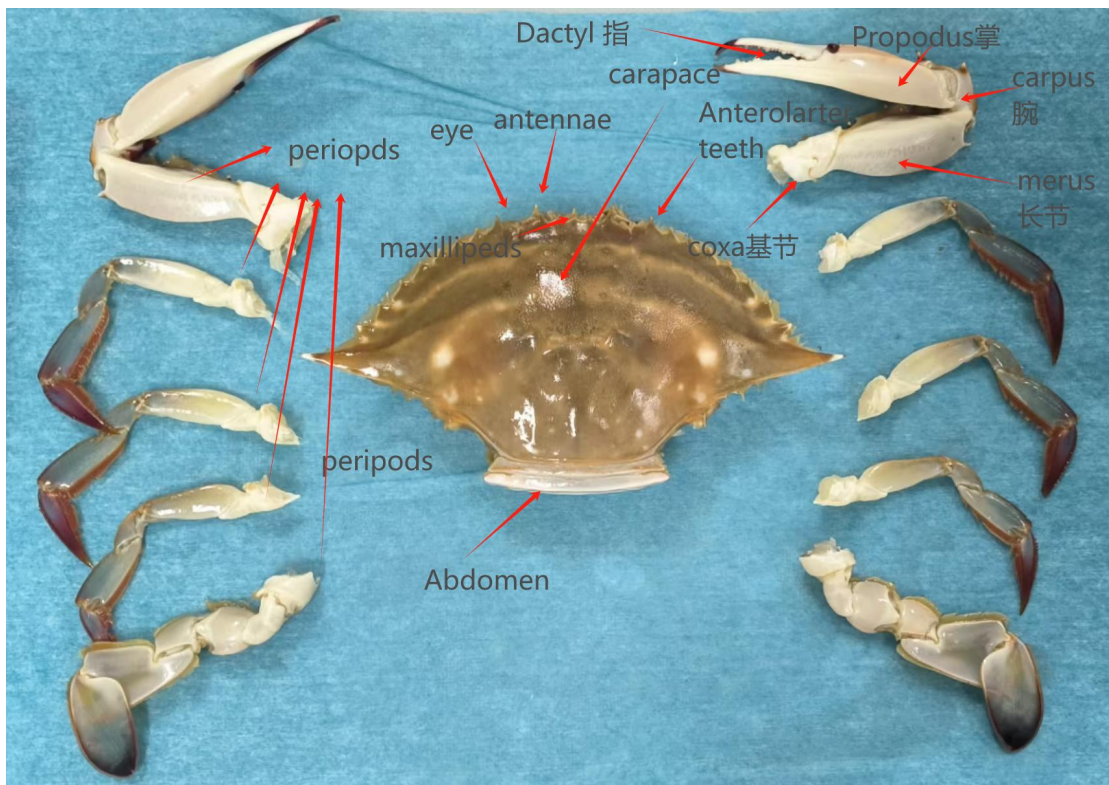
1.Select one Portunidae and put it on ice.Observe it to identify its gender and basic body structure.Measure the size and make record.

2.Homogenization and Preservation of tissue samples

- (1)Cut muscle sample and digest sample(no bigger than  $0.27\text{ cm}^3$ ,3 replicates)
- (2)Put 1.5ml centrifuge tubes,add sample into it.Add 500 $\mu\text{l}$  PBS into each tube.Homogenize it 1min with homogenizer 1min.Then add 500 $\mu\text{l}$  PBS again and homogenization 10s.
- (3)Put tubs into centrifuge,spin at  $4^{\circ}\text{C}$ ,12000g for 5min.When it finish,collected supernatant and transport it to another clear 1.5ml tube.Mark it and sent it to  $-20^{\circ}\text{C}$  Refrigerator
- (4)Clean up everything you used.Fold all materials into paper towels and put into plastic bags.Rinse and sanitize dissection equipment.

### 5. Question

- (1) take picture of your samples;identify the external structures and functions



Maxilliped: Water Filtration, Respiration, Cleaning, Defense, Feeding Assistance

Periopods: capturing food or defence

Pleopods: swimming

Eye: observe environment

Carapace: protect

Abdomen: Contains accessory limbs

(2) Dissect the organisms and Fill in the gaps for Portunidae.



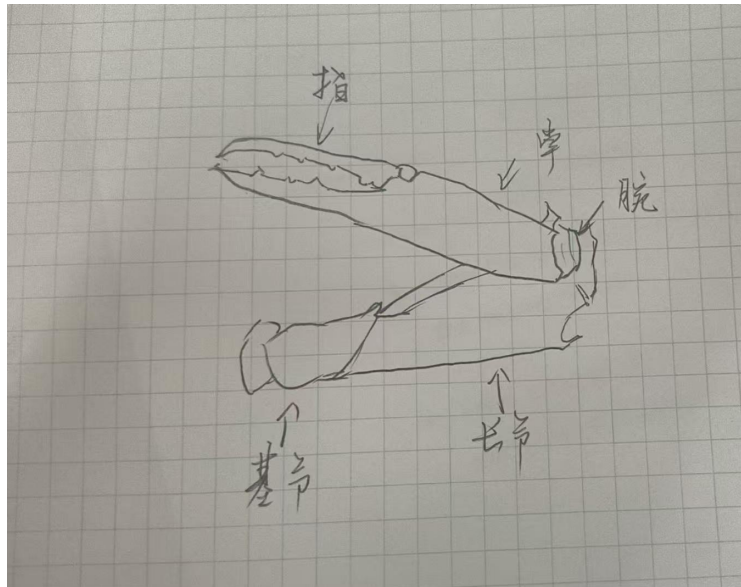
2. Dissect the organisms and Fill in the gaps for Portunidae:

|        |            |        |        |    |                                    |
|--------|------------|--------|--------|----|------------------------------------|
| Sample | Portunidae | Gender | Female | Cm | long wide high<br>13.3 x 6.2 x 5.4 |
| Size   |            |        |        |    |                                    |

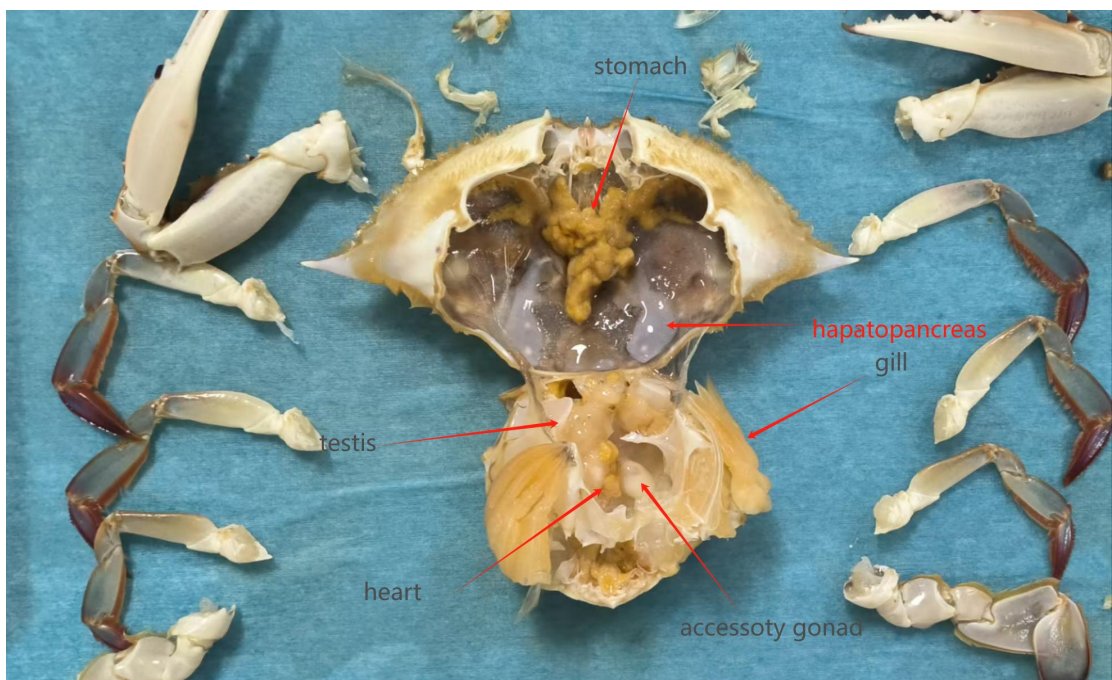


|                                                                                     |                             |                                                                                      |                          |
|-------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------|--------------------------|
|  | 1 <sup>st</sup> Antennae    |  | 1 <sup>st</sup> Periopod |
|                                                                                     | 2 <sup>nd</sup> Antennae    |                                                                                      | 2 <sup>nd</sup> Periopod |
|  | 1 <sup>st</sup> Maxilla     |                                                                                      | 3 <sup>rd</sup> Periopod |
|                                                                                     | 2 <sup>nd</sup> Maxilla     |                                                                                      | 4 <sup>th</sup> Periopod |
|                                                                                     | 1 <sup>st</sup> Maxillipede |                                                                                      | 5 <sup>th</sup> Periopod |
|                                                                                     | 2 <sup>nd</sup> Maxillipede |  | 1 <sup>st</sup> Pleopods |
|                                                                                     | 3 <sup>rd</sup> Maxillipede |                                                                                      | 2 <sup>nd</sup> Pleopods |

(3) Draw the detail of the first periopod of Portunidae and illustrate the related names



(4) Take pictures of the dissected samples and identify the main internal structures and discuss their functions.



Stomach: Digest

Intestine:Transportation, nutrient absorption

Hepatopancreas: hematopoiesis

Testis:storage of sperm

Accessory gonad: Production sperm

Heart:Blood supply

Gonopores:Intersection location

### (5) Sample quality table

| Sample number | tube quality (g) | Sample and tube<br>quality (g) | Sample quality<br>(g) |
|---------------|------------------|--------------------------------|-----------------------|
| M1            | 0.8270           | 0.9685                         | 0.1415                |
| M2            | 0.8389           | 0.9930                         | 0.1541                |
| M3            | 0.8381           | 0.9914                         | 0.1533                |
| D1            | 0.8398           | 0.9952                         | 0.1554                |
| D2            | 0.8355           | 0.9815                         | 0.1460                |
| D3            | 0.8250           | 0.9930                         | 0.1680                |

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注：不同类型的实验课对实验报告可有不同要求，各个学科的实验报告可以根据自己的学科特点做适当的调整，但上述基本内容中的第一、二、三、六条为必须填写的内容。